iSort Final Report

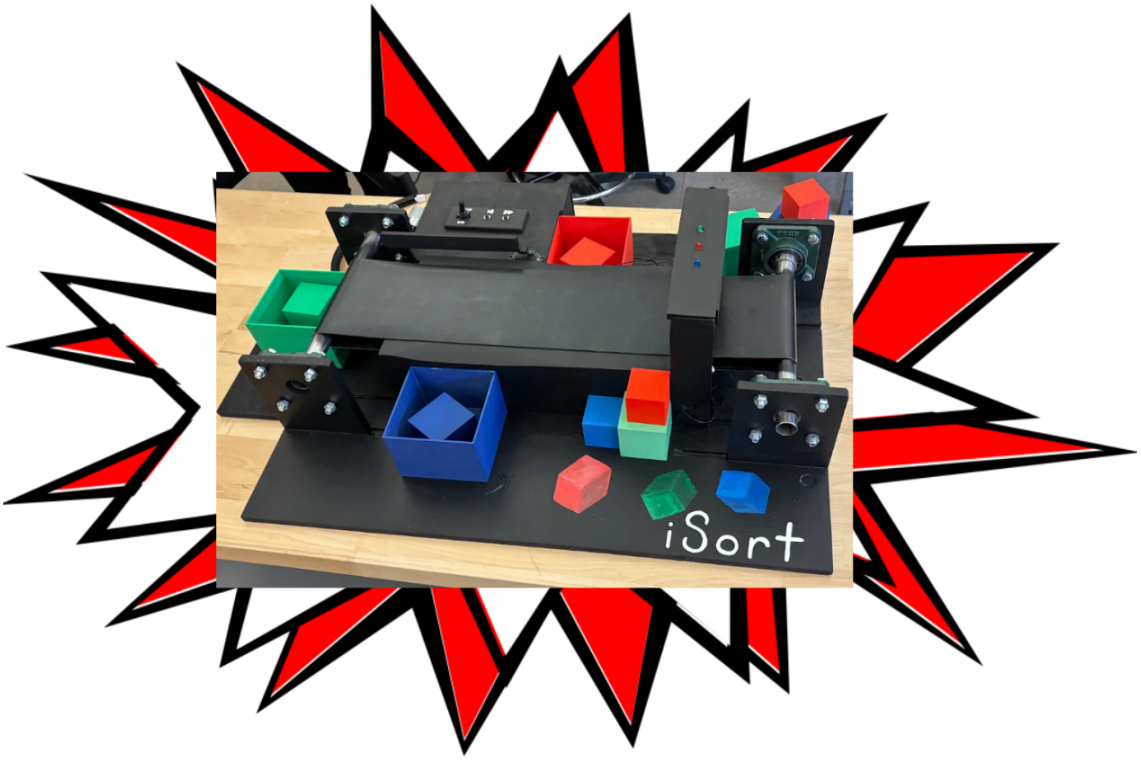
Conveyor based color sorting system

MECH 307 | Group 11

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Introduction

Mech 307 was a challenging course designed to test students' knowledge of electrical and mechanical systems and how they work together. Each team was tasked to create a project which integrated systems and satisfied requirements within 6 different categories: Output Display, Audio Output Device, Manual User Input, Automatic Sensor, Actuators, and Logic. To accomplish this goal, Group 11 devised a device known as 'The Color Sorter,' a system that would detect the color of an object and move a servo-actuated flap to a predetermined angle, allowing the object to fall into a bucket of the corresponding color.

Design Summary

The Color Sorter is a compact device designed to move objects along a conveyor belt, detect their color on a red, green, and blue scale, and direct them into matching bins. As shown in Figure 1, the system integrates mechanical and electronic components on a stable wooden platform.

Objects are transported by the (6) conveyor belt, which is driven by the (1) DC motor. The belt is supported by rollers mounted on (2) bearings. Electrical components, including the Arduino Mega, 2S LiPo battery, wiring, buck converters, and power distribution circuitry, are housed in the (3) electrical box, which serves as the central control and power hub.

Mounted on the electrical box are a main power switch, a start/pause button, and a reverse button. The (4) power plug connects the 5 V wall adapter, whose output is placed in series with the reused battery to provide approximately 12 V nominal to the system.

As an object moves along the belt, its color is detected by the (7) color sensor inside the sensing dome. An LED strip around the inside of the dome illuminates the object. The (8) indicator LEDs display the detected color. The (9) servo-arm redirects the object into one of the (5) 3D-printed collection boxes. These components provide automatic sorting, visual feedback, and manual control.

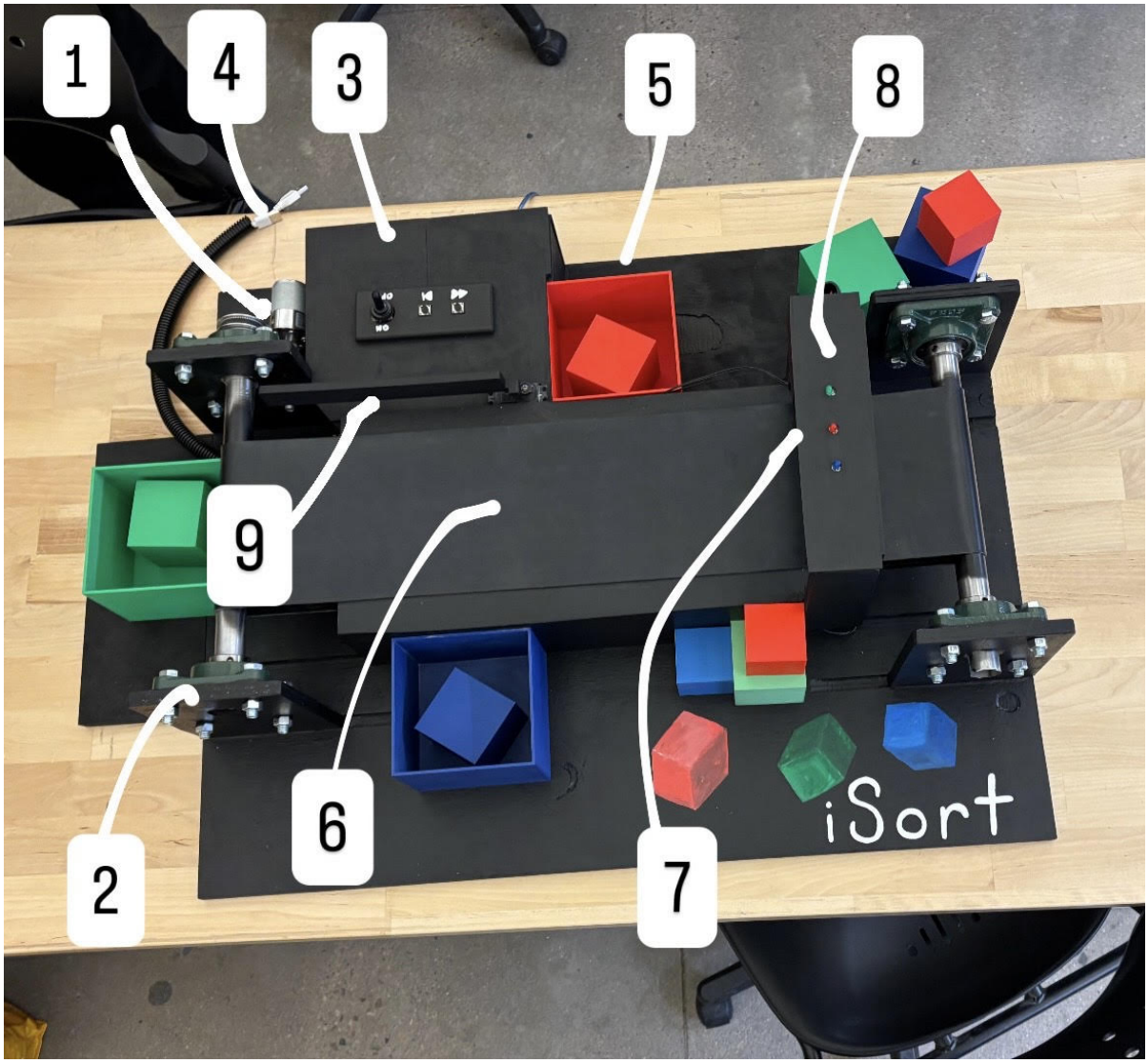


Figure 1: Labeled Photograph of Color Sorter

System Details

Mechanical System Design and Function

The mechanical structure consists of a conveyor belt mounted on a wooden frame. The belt is driven by a 12 V DC motor and supported by rollers mounted on bearings. Objects are carried toward the sensing region, where they are positioned beneath the color sensor. Once identified, a servo-driven guiding arm rotates to a predetermined angle so that the object travels into the appropriate collection box.

Electrical System and Power Supply

The electrical system is centralized within the electrical box, which houses the Arduino Mega, power distribution wiring, buck converters, and battery. The input supply combines a 2S LiPo battery rated at 7.4 V nominal with the DC output of a 5 V wall adapter connected in series. Their nominal sum is 12.4 V, treated as approximately 12 V for the project. The actual battery voltage varies with its charge level.

We used this setup because the 2S battery was available from a previous project and we wanted to avoid purchasing a larger battery. The wall adapter is hence an integral part of the main power supply. Buck converters provide a 5 V logic supply and a 6 V servo supply. The L298N motor driver controls the direction of the conveyor motor. Figure 2 summarizes the functional power distribution.

Sensors

An IR break-beam sensor detects an object and signals the controller to stop the belt for measurement. The TCS34725 sensor measures red, green, blue, and clear light channels. The LED strip around the inside of the same dome provides local illumination to make readings more consistent across rooms with different lighting conditions. The dome and strip reduce the influence of ambient light. Separate red, green, and blue LEDs show the selected color, while a buzzer signals an unrecognized object.

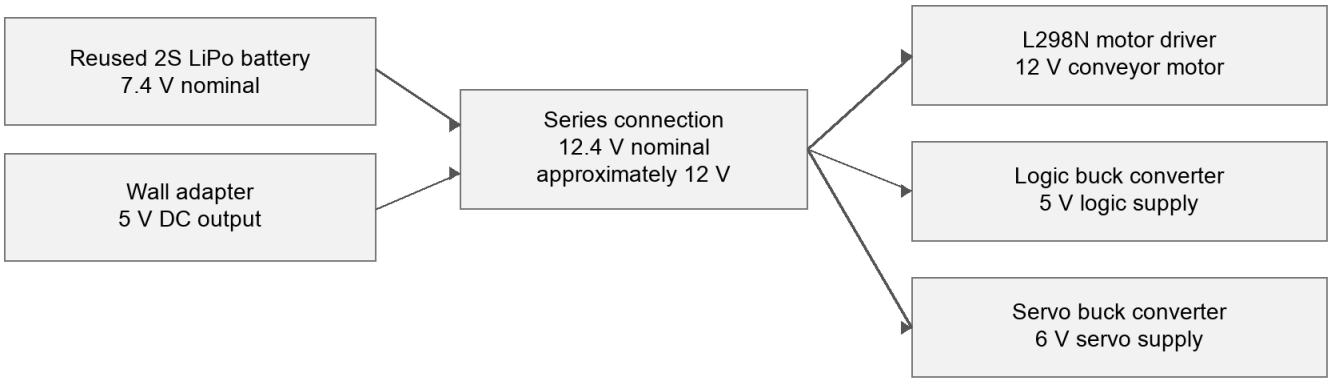


Figure 2: Functional Power Distribution

Software Logic and Control

The Arduino Mega is the sole microcontroller of the project. When the IR sensor detects an object, the belt stops and the color sensor starts a color-detection. The controller uses a gain of 4 times and an integration time of around 50 ms. We used elapsed-time checks for waiting, buzzer sounds, flap movement, and reverse operation using millis(). This was done to keep buttons responsive during every stage of operation.

Color Decisions and Outputs

The controller compares the red, green, and blue readings relative to their total signal. A color is detected only when one color makes up at least 45% of the RGB total and is higher than the next strongest color by at least 10%. These rules are based on the 3D-printed blocks and the neoprene belt we had because the color detection was never properly calibrated due to time constraints. The table below shows the general structure of the color-sensing logic (Table 1).

Reading	Decision	Output
Red dominant	Red	Red LED and flap at 145°
Green dominant	Green	Green LED and clear path at 18°
Blue dominant	Blue	Blue LED and flap at 65°
Belt **color**	Belt	Keep the belt going; no change in flap; no output
Other valid reading	Unknown	Buzzer; flap at 97°, which blocks the way; reverse the belt to reject it
Dark, saturated, or failed reading	Sensor fault	Stop conveyor; retry sensor initialization

Table 1. Color classification logic and corresponding output behavior

The belt condition is triggered when the RGB combination does not match red, green, or blue categories and red and green values each exceed 0.4 of the clear-channel reading, and blue is below 0.5 of that reading. When triggered, the belt resumes, the flap does not move, and no output function is displayed. Other combinations trigger Unknown, which moves the flap to block the way and reverses the belt for 3.5 seconds to reject the object of unknown color.

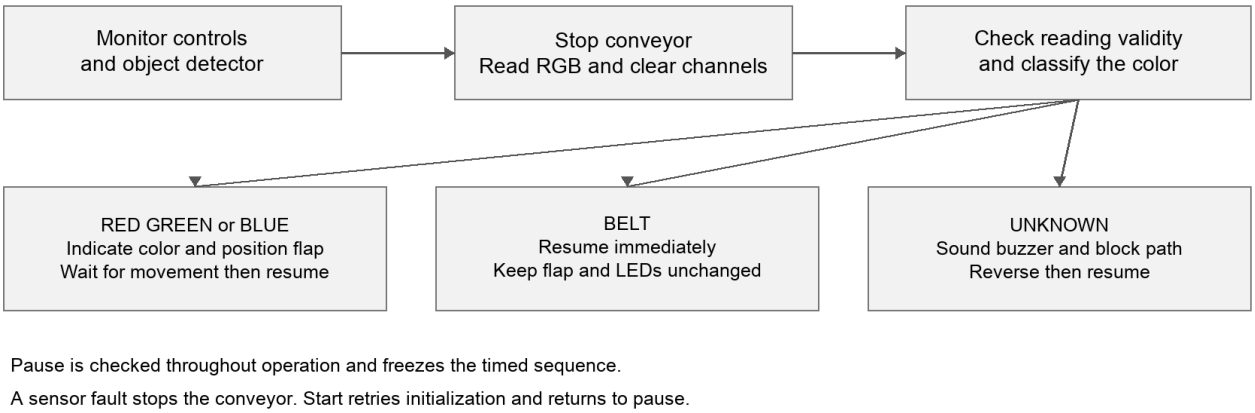


Figure 3: Control Diagram

Start/pause freezes and resumes the timed sequence. Holding reverse while paused moves the belt backward. Reversing cancels any previous color classification so that the object can be measured again.

Design Evaluation

The Color Sorter successfully integrates and completes all functional element categories, with multiple components within each group.

Output Display

Three indicator LEDs light up according to the red, green, or blue classification. The illumination strip inside the sensing dome is not an output device. It lights the measurement area to help with color detection. It does not output the result of the classification.

Audio Output Device

A buzzer sounds when an object does not fall into any of the defined categories: red, blue, green, or belt.

Manual User Input

A switch turns the machine on and off. Two buttons serve as a start/pause control and manual reverse. The reverse button works when the system is paused, which allows the user to manually change the position of the object.

Automatic Sensor

Two sensors fulfill this category. The RGB color sensor identifies the light reflected by an object, and the IR break-beam sensor detects when an object crosses its path.

Actuators Mechanisms and Hardware

A servo motor directs blocks to their corresponding bins, while a 12 V DC motor drives the conveyor. Manufactured and 3D-printed parts form the frame, collection boxes, guiding arm, and other mechanical features.

Logic Processing and Control

The controller handles object detection, motor control, color classification, inputs, and outputs.

Parts List

This section presents the specialized components used in our conveyor-based color-sorting project. It highlights the actuators, sensors, and other nonstandard hardware required for object detection, motion control, and user interaction, along with their model numbers, sources, and costs.

Table 2 below shows the list of specialized components.

Part	Purpose	Model	Source	Qty.	Cost
Color Sensor	Detects color of object	TCS34725	Amazon	1	$4.50
IR Break Beam Sensor	Detects if an object crosses its path	N/A	Nelson Mechatronics Lab via Amazon	1 set	$5.50
Servo Motor	Positions the guiding arm	MG995	Team Member - Jason	1	$0.00
12 V DC Motor	300 RPM Motor, used to turn Belt	ZGB37RG17 4i	Team Member - Bella	1	$0.00
Motor Driver	Controls speed and direction of motor	L298N	Team Member - Bella	1	$0.00
Logic Buck Converter	Converts 12 V to 5V	LM2596	Nelson Mechatronics Lab via Amazon	1	$1.30
Servo Buck Converter	Converts 12 V to 6V	N/A	Nelson Mechatronics Lab via Amazon	1	$13.99
Fuse Holders	Hold fuses for circuit protection	N/A	Nelson Mechatronics Lab via Amazon	2	$2.63
Toggle Switch	Switches power on and off	KN3C-101	Nelson Mechatronics Lab via Amazon	1	$2.50

Table 3 below lists the mechanical components used for the project.

Part	Purpose	Source	Qty.	Cost
Neoprene Rubber Sheet	Used for the belt	Amazon	1	$24.99
Pulley	Attached to motor to rotate belt	Amazon	1	$14.21
Bearings	Secured rods which held belt in place	Amazon	4	$27.78
Wood	Used for the frame	Home Depot	2 ft x 4 ft Plywood	$26.95
Metal Curtain Rods	Used to hold the belt	Team Member - Kamoliddin	2	$0.00
LED Strip	Illuminates the object inside the sensing dome	Amazon	1 strip	$6.95
Wire Tubing	Used to cover up wires	Amazon	1 ½” diameter, 10 ft long tube	$5.05

Recorded Project Spending

Category	Recorded cost
Specialized components	$30.42
Building materials	$105.93
Total listed project spending	$136.35

Components supplied by team members are recorded at zero cost. Reusing the motor, motor driver, servo, rods, and battery from previous projects reduced the amount of new hardware we needed to purchase.

Lessons Learned

This project was a challenging but rewarding experience and included significant problems in mechanical assembly, wiring, and sensor operation. At first, the color sensor would not turn on, and when it did, it read the wrong color. Improved wire connections restored operation, and changes to the classification conditions helped the team distinguish red, green, and blue objects.

Another problem was the small ground wire associated with the wall power connection. It broke multiple times and required repeated stripping and resoldering. An extension cord reduced the need to stretch the connection toward the outlet and lessened stress on the wire. Soldering smaller wires to the small lead also worked better than trying to join it directly to a larger stranded wire.

The most significant difficulty occurred the night before the project was due, when the Arduino controlling the system failed and had to be replaced. This led to upload problems and wiring errors, and the color sensor was damaged and required replacement. A complete rewire of the system identified a bad buck converter as the underlying problem. Resolving the issue required time, cooperation, and careful checks across the entire electrical system.

Recommendations for Future Students

- Keep wiring neat and tidy. The more wires you have, the more confusing it can be to know which wires go where, and when one may be plugged into the wrong area, which could throw off the entire project.
- Keep an iterative process. With many components having to all work together, starting with implementing and coding one part, and making sure it works, before continuing on to the next part, is important in making sure bugs in the code or faulty wiring can be found and fixed before they cause bigger problems.

- Maintain exceptional group communication. This project is impossible to do alone, and the only way success will be found is by working together each step of the way and utilizing each group member's strengths to get each part of the project done efficiently and effectively.
- Always have a backup plan. Mechanisms can often not work as intended or break, and it’s very important to have additional parts or another part to implement in case something goes wrong.
- Start early. You will run into problems, and not everything will go smoothly. Allow more time than you think you will need to assemble and test your project.

Future Work

The main priorities are replacing the combined power source with a single source and calibrating the color sensor using real data to improve the classification logic. Then, experimental validation for color sorting speed and accuracy should be implemented.